* find out the list of companies which are conducting the placement ~~rights~~ drive using DSA course

→ findout the syllabus of gate for DSA.

→ find out set of I & O for the following.

① Electricity bill

② Bus ticket

③ Message to a friend

→ today when you started from ~~home~~ hostel to college what are the various data structures you identified in your surroundings.

gate syllabus

1. Programming & Data Structures
   - → C programming basics
   - → Recursion
   - → Arrays
   - → Stacks
   - → Queue
   - → Linked Lists
   - → Trees
   - → Binary Search Trees
   - → Binary Heaps
   - → Graphs
2. Algorithms
   - → searching - Linear search
     - Binary search
   - → sorting - Bubble sort, selection sort, Insertion sort, Merge sort, Quick sort, Heap sort
   - → Hashing
   - → Time and space complexity - Big-O(O)
     - Omega(Ω)
     - Theta(Θ)
   - → Algorithm Design Techniques - Divide and conquer
     - Greedy Algorithms
     - Dynamic programming
   - → Graph Algorithms - BFS
     - DFS
     - Minimum spanning tree
     - Shortest path.

Tier 1 (very high DSA)

- Google
- Microsoft
- Amazon
- Meta
- Apple
- NVIDIA
- Uber
- Atlassian
- Adobe
- Salesforce

Tier 2 (strong DSA)

- Flipkart
- Walmart
- Intuit
- Oracle
- Paypal
- visa
- American Express
- Goldman Sachs
- Morgan Stanley
- D.E.Shaw

Indian product companies

- Phonepe
- Razorpay
- Meesho
- Groww
- CRED
- Freshworks
- Zoho
- Juspay
- Swiggy
- Zomato

Service Based companies (Basic to moderate DSA)

- TCS
- Infosys
- wipro
- Cognizant
- Accenture
- Capgemini
- HCLTech
- Tech Mahindra
- IBM

③ Electricity bill:

Input:

- Consumer Number
- Customer Name
- Previous Meter Reading
- Current Meter Reading
- Rate per unit
- Fixed charge

Output:

- Units Consumed
- Electricity Bill Amount
- Due date
- payment status

2) Bus ticketing

Input:
- passenger Name
- source
- Destination
- Date
- Bus Type
- Number of Tickets.

Output:
- Ticket Number
- Passenger Name
- Seat Number
- Fare
- Bus Number
- Journey Date

3) Sending a message to friend.

Input
- Sender Name
- Receiver Name
- Message
- Date
- Time

Output:
- Receiver
- Message
- Date & Time
- status.

4) Mess token — object
- plates — array
- classroom benches — array
- Books — array
- Subjects — array
- attendence register — array

→ Create a structure variable student and estimate the size of the structure manually then write a C-program to find size of a structure using size of operator now observe both the outputs. Are they same or different? Justify your answer.

→

```
#include<stdio.h>
struct student
{
    int rollno;
    char grade;
    float marks;
};
int main()
{
    struct student s;
    printf("size of structure = %zu bytes\n", sizeof(s));
    return 0;
}
```

Manual

int - 4
char - 1
float - 4
} total = 9bytes

Program o/p:

Size of structure = 12bytes

No, they are different. Because of structure padding (alignment). The compiler inserts extra unused bytes b/w members or at the end of the structure so that data is aligned in memory for faster access by the CPU.

→ ① Open amazon website and find out 10 different functions.

→ ② Return 1 or return 0 means what, the main returns to whom.

①A: (1) Search products (2) Product categories
(3) Add to cart (4) Buy Now (5) Wishlist
(6) Customer Reviews & Ratings (7) Order Tracking
(8) Filter and sort (9) User login (10) Payment

②A) It return the value to operating system (os).

return 0: program executed successfully.

return 1: program did not complete successfully.
It contains 'general' error / failure.

→ Algorithm

Q: In java every object has hash code findout what is hash code and why it is being generated in java.

what is hash code? - It is an integer value that represents an object. It is mainly used to identify objects efficiently in hash-based collections such as: HashMap, HashSet, HashTable

why it is generated? to improve searching speed, helps compare objects efficiently, determine the bucket.

eg:

```
class Demo{
    public static void main(String[] args){
        String s = "Hello";
        System.out.println(s.hashCode());
    }
}
```

o/p:

6960650.

→ H/W

compute that time complexity using frequency count for the following:

① `for(i=1; i<=10; i=i+2)` — 1+1+2 — 1+11+20 — 32

```
{ i = i&10;
  printf("%d", i);
}
```

`i = i&10;` — 1+1 — 20 — 20 (10×2)

`printf("%d", i);` — 1 — 10 — 10

62

② `i = 1;` — 1 — 1 — 1

`while (i<=50);` — 1 — 1 — 1

2

```
{ printf("hello");
}
```

③ `for(i=1; i<=20; i++);` — 1+1+1 — 3 — 3

④ `for(i=1; i<=n; i=i++)` — 1+1+2 — 1+(n+1)+2n — 3n+2

```
{
  printf("hello");
  printf("world");
  printf("i=%d", i);
}
```

`printf("hello");` — 1 — n — n

`printf("world");` — 1 — n — n

`printf("i=%d", i);` — 1 — n — n

6n+2